Training load and injury: part 2. Questionable research practices hijack the truth and mislead well-intentioned clinicians

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Abstract

Background:
In this clinical commentary, we highlight issues related to conceptual foundations and methods used in the training load-injury research. We focus on sources of degrees of freedom that can favour questionable research practices such as p-hacking and HARKing (Hypothesizing After the Results are Known), which can undermine the trustworthiness of research findings.
Clinical question:
Is the methodological rigour of studies in the training load-injury field sufficient to inform training-related decisions in clinical practice?
Key findings:
The absence of a clear conceptual framework, causal structure, and reliable methods can promote questionable research practices, selective reporting and confirmation bias. The fact that well accepted training principles (e.g. overload progression) are in line with some study findings may simply be a consequence of confirmation bias, resulting from cherry picking and emphasizing results that align with popular beliefs. Identifying evidence-based practical applications, grounded in high-quality research is not currently possible. The strongest recommendation we can make for the clinician is grounded in common sense "do not train too much too soon" – not because it has been confirmed by studies, but because it reflects accepted generic training principles.
Clinical application:
The training load-injury research field has fundamental conceptual and methodological weaknesses. Therefore, making decisions about planning and modifying training programs for injury reduction in clinical practice, based on available studies, is premature. Clinicians should continue to rely on best practice, experience and well-known training principles, and consider the potential influence of contextual factors when planning and monitoring training loads.

Key words: training load, injury, conceptual model, research methods, risk of bias

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Introduction: a self-fulfilling prophecy

"We must be careful not to believe things simply because we want them to be true, no one can fool you as easily as you can fool yourself"

Richard Feymann

It is tempting to believe a theory when it appears reasonable and it fits with one's beliefs. For example, the theory that training "too much" or "too little", or "too much too soon" might cause sports injury seems biologically plausible and aligns well with training dogma. Common beliefs about how training load affects the chances of the athlete getting injured have been strongly shaped by a combination of old and deeply-held training principles, best practice and common sense. Progressive overload, as well as the 'danger' of excessive training (overtraining or non-functional overreaching) are well-recognized training principles. The idea that excessive training can increase injury risk can be traced back to early 1990s.[44]

Popular beliefs about training load and injury obfuscate important concepts and methodological issues. Unfortunately, these issues have not been well accounted for in previous research. Our concern is that the training load and injury research field is dominated by well-intentioned, yet potentially misleading, recommendations for clinical practice.

Collecting injury and training load data is now considered best practice in sport.[14] An unfortunate consequence of such a data rich environment is that researchers studying the relationship between training load and injury may retrospectively select from a convenient sample of available data rather than developing clear and well-defined research questions *before* collecting data. There are many studies reporting associations between training load and injury. The problem is, without a clear conceptual framework, it is easier for confirmation bias or selective reporting to creep in and hijack the truth.

Clinicians are exposed to an ocean of information. One challenge for clinicians is to unravel the "signal" from the "noise" and identify relevant findings that can be confidently applied in practice. Most clinicians do not possess the ability to identify studies at high risk of bias, as this requires in depth research methods training. The aims of our clinical commentary are to (1) help clinicians identify some of the methodological weaknesses in training–load injury studies, and (2) demonstrate why clinicians should exercise caution when applying findings from these studies to their practice.

Clinical question

Is the methodological rigour of studies investigating relationships between training load and injury sufficient to inform training load-related decisions in clinical practice? We have highlighted why it is important to critically examine the strength of evidence supporting the claims made in several training load-injury studies.[36] In this clinical commentary , we focus on sources of degrees of freedom that can favour questionable research practices (including p-hacking and HARK-ing (Hypothesizing After the Results are Known)) that can impact on the trustworthiness of research findings and the application of training load metrics in clinical practice.

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A robust conceptual framework: the first step in designing quality research to help clinicians and athletes

Developing a conceptual framework is an essential early step when designing research to inform clinical practice. If properly developed, a conceptual framework can guide researchers' hypotheses and specific research questions, and provide a *frame* for analyzing data and interpreting the results.[34 54] HARK-ing is a bit like doing research in reverse. When (well-intentioned) researchers do not develop a clear conceptual framework before collecting research data, they risk fashioning a hypothesis to suit their research results, and the temptation to conduct (unplanned) analyses until they find something "significant" (p-hacking).

There have been attempts within the training load injury field to provide generic concept maps and models.[62] These can help clinicians understand the multifactorial nature of injuries and contextual factors that influence injury, but they lack detail to define precise research questions and/or select specific training load metrics. This necessitates the use of plausible biological and physiological rationale to select specific training load measures of exposure as potential prognostic factors for certain injuries. The conceptual framework proposed by Bertelsen et al.,[6] is the most appropriate model of the causes of running-related injury (Figure 1).[41] While one can argue the suitability of specific proxy measures (e.g. the use of ratios, or 7 and 28 days cumulative load), we applaud the authors for transparently presenting their research assumptions, and encourage researchers to consider using the framework[6] as a starting point for new projects.[42]

Figure 1*

Conceptual frameworks must be verified (or disproven) through specific study of the hypothesized relationships between variables in the framework. If the hypotheses are confirmed by original studies, the model can be accepted as a reasonable explanation of the relationships. If the hypotheses are not confirmed, the researchers must go back to the drawing board and re-think their hypotheses. Research must challenge the assumptions and logic inherent to the framework to test its strength. Without a conceptual framework and predefined hypothesis about the relationship between variables, there is a risk of confirmation bias as the researchers may attempt to assign meaning to results.

Without understanding and testing underlying etiology using a framework, it is impossible - irrespective of statistical approach - to accurately interpret research results. A satirical study in American Football[55] illustrated that the risk of concussion was linked to the team logo: Teams with animal logos were protected from concussion. Should teams consider changing their logos or implementing 'protective' animal stickers on their equipment to reduce the concussion risk?

We expect most clinicians would agree that changing a logo is unlikely to change concussion risk. This article illustrates how, in the absence of a strong physiological rationale (i.e. framework), coincidental links may be misconstrued as clinical (or practical) recommendations. Our concern is that in absence of an established

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conceptual framework, the link between training load and sports injury might be misleading clinicians in sports science and medicine practice.

A fishing expedition: p-hacking

Many studies have reported associations between training load and sports injuries. However, the results are inconsistent and often confusing.[17] We summarised the results (odd ratio (OR), relative risk (RR), and injury risk) of studies in soccer that calculated the ACWR from in-season session rating of perceived exertion (sRPE) or Global Positioning System (GPS) measures for non-contact injuries (Figure 2, Figure 3). Some studies have reported a relationship between *high* ACWR and injury, some have reported a relationship between *low* ACWR and injury, and others have reported no relationship between ACWR and injury.

Using session RPE, three studies[19 46 47] reported increased injury risk when the ACWR was high (Figure 2).[4] In one study,[39] the results were the opposite: a lower injury risk when the ACWR was high (Figure 2). In another study,[16] there was no relationship between ACWR and injury injury 8 out of of 9 comparisons. For GPS-derived measures (figure 3), in one study there was no relationship between ACWR and injury risk in 4 out of 5 comparisons,[10] elevated injury risk at high ACWR for total distance and accelerations but not for high speed distance and decelerations,[9] and higher odd ratios for high-speed running, but not accelerations and decelerations.[39] Confused? It is difficult to reconcile the inconsistencies to a cogent statement about the relationship between training load and injury. And these inconsistencies are not limited to studies in soccer, or the ACWR metric.

When one considers all of the relationships between all of the variables in all of the studies, what stands out is that some results might have been emphasized and others received little (or no) discussion – in research methods, this is referred to as selective reporting bias. The researcher emphasises the results that fit with his or her preconceptions, or the common beliefs about training load. For example, emphasizing three studies that supported a protective effect of moderate ACWR, and omitting other studies that did not show the same trend.[21] Selective reporting (and discussion) of study findings misleads clinicians and researchers.

***** Figure 2 *****

***** Figure 3 *****

Inconsistency between and within studies in the training load–injury field, is often justified by the multifactorial nature of injury etiology. While this is true and models accounting for this complexity have been proposed,[8 32 33] the multifactorial nature of injury cannot be used as an excuse to ignore inconsistency. Instead, the complexity must be overcome with robust studies. A small number of training load metrics cannot adequately explain injury risk in sport.

According to the Grading of Recommendations Assessment, Development and Evaluation (GRADE) guidelines, inconsistency in the results of similar studies is grounds to downgrade the credibility of evidence.[22] We recommend that researchers

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and practitioners examine all the results of the studies, avoiding or recognizing selective discussion. Be on the alert for implausible and inconsistent findings (also called unexpected associations), which may suggest associations due to chance, misclassification of the predictor, selection bias, mixing of effects (confounding), intervention effects and heterogeneity.[53]

The decisions researchers make affect the likelihood of HARK-ing and p-hacking: threats to the credibility of results in the training load–injury field

We believe the training load–injury field is at high risk of 'data fishing' – where researchers go searching (consciously or unconsciously) for answers in the data to confirm a relationship between training load and injury. Clinicians trying to apply research findings in practice may not be aware of all of the choices researchers must make when conducting a study, and how each choice might influence the results.

Sometimes, the choices researchers make might lead to false discoveries that favour their beliefs (confirmation bias). In this section, we highlight 8 methods issues (choices to be made by the researcher) that can increase the risk of p-hacking and HARKing ('data fishing').

i) Measures of exposure and ratios
In the absence of a strong a-priori conceptual framework, researchers are free to select the training load metric(s) they wish to include in a statistical model after the data are collected. Because there are so many different measures of training load (exposure),[35] different researchers might make different choices. How does one interpret different results and different metrics, especially when there is a high chance of false discoveries? Different training load metrics may also have different relationships to injury risk.[9 10 39]

Common measures of exposure (e.g. acute and chronic training load, and their ratio) lack conceptual and computational validity.[38 56] Ratio measures of training load are common in the training load–injury field. However, the dangers of using ratios (as normalization method or dependent/independent variables) have previously been described, and the pitfalls well documented.[3 15 58] When clinicians and researchers ignore warnings about using ratios as the main training load measure, they risk falsly concluding that the quantity of training load causes injury.

ii) Training load measures
Measures of exposure are calculated using various training load metrics. However, training load can be assessed using different methods and devices, which further complicates the selection of suitable measures. Again, a conceptual framework should provide guidance when selecting appropriate training load measures. Although studies have used sRPE, GPS, and inertial sensors, each of these measure different training load constructs.[9 10] Therefore, combining injury rates from different sports and training load measures[7] is inappropriate.

iii) Time windows
Without a conceptual framework it is not possible to justify what is an appropriate time window within which to measure *acute* load, and what is an appropriate time window within which to measure *chronic* load. The original Banister model cannot

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and should not be used to derive these time windows because the time decays used in the model are conceptually very different than weekly average training load. The solution to trial several windows concurrently (to find the best model)[11] is prone to bias, multiple testing concerns and overfitting.

iv) Time lags
Typical time lags between injury registration and the acute or chronic load typically range from few days to one week (subsequent week injuries)[11 29 31 40] but longer lags have been used.[40 59] There is no reason to expect that the training completed on the day or days before an injury would not affect injury risk. Researchers do not usually explain why they chose a particular time lag over another time lag, raising suspicion of p-hacking.

v) Discretization and reference category
Most studies lump training load measures and ratios into categories (Figure 2, Figure 3). Such an approach has serious limitations,[1 5 12] including that the number of categories can influence the results and subsequent interpretation. Using categories exacerbates the risks and consequences of sparse data bias (some categories have many data points, some categories have very few data points). Results from studies using different categories and references should not be compared.

vi) Statistical analysis
Most studies examining the association between training load and injury have used inadequate statistical analyses,[49 61] including approaches that cannot account for time-varying variables, recurrent events or repeated measures.[29-31] The challenges and solutions for more appropriate analysis have been provided, but are rarely followed.[2 49 50 52] This is a problem because inappropriate analysis can produce unreliable and biased results.[57]

vii) Missing data
Missing training load data are common and sometimes unavoidable. Most studies in the training load–injury field do not descibe how missing data were handled. For example, were data imputed? What were the assumptions the researchers made about the missing data?[24 57] Imputation can have an effect on the results. Sensitivity analyses should be performed to show the effects of the methods, as recommended by international guidelines.[13]

viii) Injury definitions
The injury definition a researcher chooses is important. Some definitions are very broad (e.g. all complaints); others are narrow (e.g. serious, non-contact, time loss injuries). When researchers do not provide an appropriate rationale, grounded in a solid theoretical framework and etiology model, to justify their choice, it is difficult for readers to interpret results and compare studies. [23]

Correlation does not equal causation

The dominant narrative of the training load–injury field is that by manipulating training load, one can alter the probability of future injuries. However, this assumes a cause-effect relationship (changing training load causes an injury to occur or not to

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occur) that has never been examined using appropriate methods required to make causal inferences.[20 26 51 52]

Although the field acknowledged that association is not prediction,[19 25 28 48] (associations between training load metrics and injury do not automatically imply that training load metric can predict injury occurrence) – it is important to recognize that neither associations nor predictors can be automatically used to make causal inferences if this was not the original aim.[27] Associations can be descriptive, predictive or used to estimate causal effects. However, if no causal relationship has been estimated, any practical applications regarding the manipulation of the prognostic factor/predictor to alter the probability of an event is an over interpretation and a speculation (and should be declared as such).

Changing a risk factor like a training load metric, if the metric does not have a causal relationship to the outcome or event, cannot modify the risk of an event occuring. Clinicians should be aware that, regardless the methodological approach, if the a study did not explicitly estimate causal effects (using appropriate methods), it is difficult to know whether changing training load causes injury. Ultimately, it should not be claimed that the intervention is "evidence-based" as often stated.

Summary

As published research in the training load-injury field has proliferated, clinicians might have been lulled into a false sense of security – accepting that training "too much, too soon" causes injuries and diligently adopting new training load metrics in hope of reducing injury risk. However, when one looks carefully at the methodological limitations and inconsistencies in previous research, evidence supporting these beliefs is not as strong as one might have expected (Table 1).

***** Table 1 *****

Recommendations for clinical practice

Given the research limitations, we encourage clinicians to follow well-established training principles.[43 45] One key principle is load progression. While some might believe that the current influx of studies have increased the attention to "correct handling " of training load, this seems to be another bias. The overload progression concept has been well-known among coaches, fitness trainers and sport scientists for at least the past half-century (training and periodization principles); but we concede this may not be the case for clinicians who may not be as familiar with athletic training methods. Stronger multidisciplinary collaboration may help when making decisions about future training. We recommend clinicians work together with the various support staff of athletes/teams so that any specific knowledge and expertise is shared.

When reading research in the training load–injury field, be on the look out for inconsistent results ("consistent" associations but in any direction is not a consistent finding) and different analysis methods that are not well justified (e.g. computational manipulations of the same prognostic factors, data trimming, categorizations, etc). Consider whether the results make sense in the practical context. For example,

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immediately after a recovery or a tapering week, would one expect athletes are at higher risk of injuries? All of these could be signs something is wrong and suggest caution when applying the results to clinical practice.

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Figure and table legends

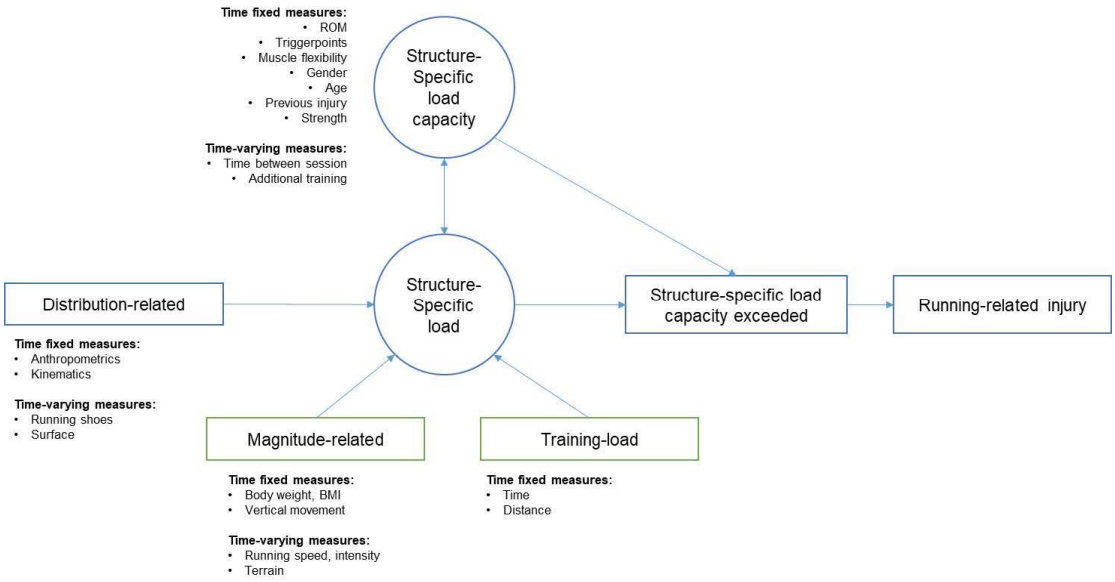

Figure 1. Simplified directed acyclic graph-inspired approach[41] to visualise the causal relationship between structure-specific load and structure-specific load and running-related injury.

Figure 2. Graphical representation of results from studies examining the association between Acute:Chronic Workload Ratio (ACWR) calculated using the session Rating of Perceived Exertion (sRPE) and injuries in soccer.
Legend: The rectangular boxes represent the ACWR categories used in the studies. When there is no rectangle, that category range was not used or reported in the results. Boxes with 'Ref' and ref2' indicate the reference category used in the statistical analysis The reference categories are shown with the exception of Bowen et al.[10] where we reported what they defined as injury risks from their tables (the values reported are rates). The studies by Ehrmann et al. (2016)[18] and Watson et al. (2017)[60]

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were excluded from figures 2 and 3 as they did not use categories. The numbers inside the rectangular boxes refer to the injury risk or rate (i.e. RR, OR) reported in the studies. The grey-shaded area represents the ACWR range (0.8 to 1.3) that is claimed to correspond to a sweet spot (lower injury risk or rate). Numbers in boxes with bold font represent measures of association not overlapping 1; * refers to associations likely or very likely, based on magnitude based inference; # refers directly to the study of Bowen who reported injury risks but the values actually corresponded to injury rate. RR, relative risk; OR, odd ratio.

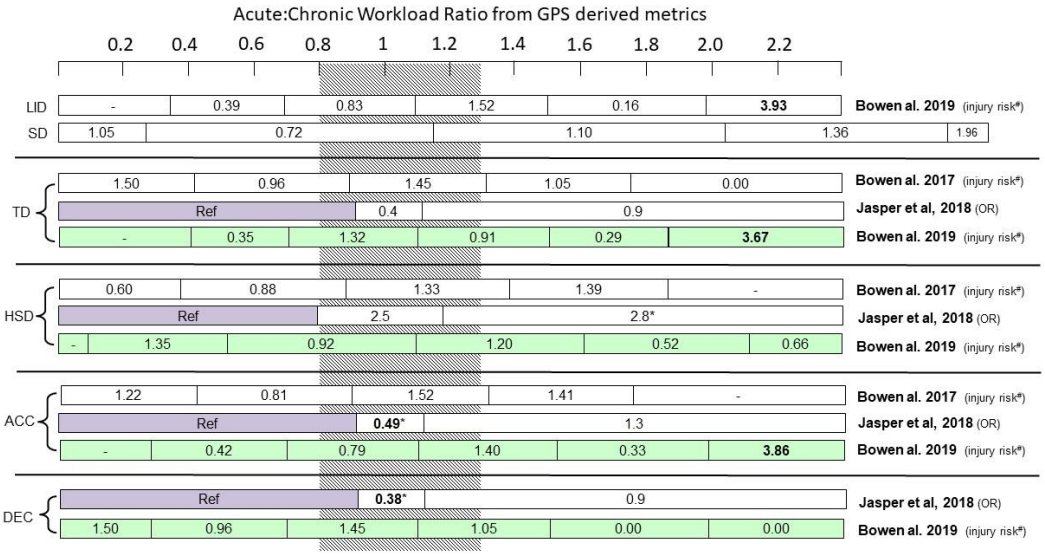

Figure 3. Graphical representation of results from studies examining the association between acute:chronic workload ratio (ACWR) calculated using Global Positioning System (GPS) and injuries in soccer.
Legend: The rectangular boxes represent the ACWR categories used in the studies. When there is no rectangle, that category range was not used or reported in the results. Boxes with 'Ref' and ref2' indicate the reference category used in the statistical analysis. The reference categories are shown with the exception of Bowen et al.[10] where we reported what they defined as injury risks from their tables (the values reported are rates). The studies by Ehrmann et al. (2016)[18] and Watson et al. (2017)[60] were excluded from figures 2 and 3 as they did not use categories. The numbers inside the rectangular boxes refer to the injury risk or rate (i.e. RR, OR) reported in the studies. The grey-shaded area represents the ACWR range (0.8 to 1.3) that is claimed to correspond to a sweet spot (lower injury risk or rate). Numbers in boxes with bold font represent measures of association not overlapping 1; * refers to associations likely or very likely, based on magnitude based inference; # refers directly to the study of Bowen who reported injury risks but the values actually corresponded to injury rate. RR, relative risk; OR, odd ratio; LID, low-intensity distance; SD, sprint distance; TD, total distance; HSD, high-speed distance; ACC, accelerations; DEC, decelerations; RR, relative risk; OR, odd ratio.

Table 1. List of the most common variations and combinations of features and outcomes used in training load-injury research.

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Table 1. List of the most common variations and combinations of features and outcomes used in training load-injury research.

Features	Outcomes (injuries)
Chronic load	*Definitions*
1. One week	1. Match loss
2. Two weeks	2. Match and training time loss
3. Three weeks	*3.* Complains, no time loss
4. Four weeks	4. Modified training
5. k days/weeks #	*Injury types*
Acute load	5. Contact
6. One week	6. Non-contact
7. k days #	7. Both contact and non-contact
Acute load calculation in relation to the injury day	*Collection*
8. Load of the same week of the injury	8. Self-reported
9. Load starting the day before the injury	9. Medical staff
10. Load starting the day of the injury	*Location*
11. Load starting the week preceding the injury	10. Lower body
TL variation metrics	11. Both lower and upper body
12. 1-wk acute/4-wk chronic (ACWR)	
13. k days or weeks acute/k days or weeks chronic #	
14. Week to week variation (relative)	
15. Week to week variation (absolute)	
16. Monotony	
17. Strain	
Computations	
18. Rolling average	
19. Accumulated (sum)	
20. Exponentially weighted moving average	
21. Coupled ($W_{-1}/W_{-1 \text{ to } -4}$)	
22. Uncoupled ($W_{-1}/W_{-2 \text{ to } -5}$)	
Categories	
23. Two (median split)	
24. Three (low, mod, high)	
25. Four (low, mod-low, mod-high, high)	
26. Five (very low, low, mod, high, very high)	
27. Six (very low, low, low to mod, mod to high, high, very high)	
28. Seven (very low, low, mod low, mod, mod high, high, very high)	
29. > seven	
Reference categories	
30. One of the categories (lower, middle, higher)	
31. All the above	
Categories determination	
32. z-scores	
33. Absolute	
34. Percentiles	
35. Arbitrary cut-points (e.g. 0.5, 0.75, 1.0, etc)	
Combinations	
36. Acute only	
37. Chronic only	
38. Variations only	
39. Low chronic vs ACWR/weekly changes	
40. High Chronic vs ACWR/weekly changes	
Training load indicators	
41. Session RPE global	
42. Session RPE on leg	
43. Ball bowled	
44. Total distance	
45. Low intensity running distance*	

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46. Moderate intensity running distance*
47. High speed running distance*
48. Very high speed running distance*
49. Sprinting*
50. Accelerations*
51. Decelerations*
52. Player load
53. Distance-load

*, various cut-off values; #, various combinations; W, week

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